

# Report - Effectiveness of isolation and contact tracing to control Bundibugyo virus (Ebola disease) outbreak in the Democratic Republic of the Congo

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## 1 Introduction

As of 24 May 2026 there were 906 cumulative suspected cases and 105 confirmed cases of Ebola virus disease caused by the Bundibugyo virus in the ongoing epidemic (Institut National de Santé Publique de la République Démocratique du Congo, 2026). The outbreak is primarily in the eastern region of the Democratic Republic of the Congo (DRC), mostly in the Ituri Province, but with confirmed cases in North Kivu and South Kivu (Institut National de Santé Publique de la République Démocratique du Congo, 2026). There are five confirmed cases in Uganda, three imported from the DRC and two secondary cases among their contacts (Reuters, 2026). On 17 May 2026 the World Health Organization (WHO) declared the Bundibugyo virus outbreak a public health emergency of international concern (PHEIC) (World Health Organization, 2026).

There is likely an underascertainment of the total cases involved in the Bundibugyo virus outbreak. A report from Imperial College London estimated the total outbreak size on 20 May 2026 was likely 400–900 cases, with a possibility of more than 1,000 cases, when reported cases were ?? (McCabe et al., 2026). A second analysis of the true outbreak size used a joint modelling approach found the most likely total number of Ebola cases on ?? May to be 930 (90% CrI 483–2234) (Abbott, 2026).

Genetic evidence points to a single spillover event with an outbreak start date around 11 April 2026 (Virological.org, 2026).

Bundibugyo ebolavirus has no licensed vaccine or therapeutic, unlike Zaire ebolavirus, for which both a vaccine (Ervebo) and monoclonal antibody therapies (Inmazeb and Ebanga) have been approved (Coalition for Epidemic Preparedness Innovations, 2026). With strong evidence of epidemic growth from increasing suspected cases (Institut National de Santé Publique de la République Démocratique du Congo, 2026) ( $R_t > 1$ ), Public Health and Social Measures (PHSMs), a type of non-pharmaceutical intervention, are required to bring the outbreak under control. Isolation of cases that test positive for Ebola, and tracing of their contacts is a standard response strategy for Ebola outbreaks (World Health Organization, 2015).

Here we test the effectiveness and feasibility of controlling the ongoing Bundibugyo virus outbreak in the DRC using a mathematical model. We simulate the Bundibugyo outbreak, assuming a single spillover event into the human population, and no PHSMs from the estimated outbreak start date to the alert of the public health emergency and activation of response

efforts. In the model isolation and contact tracing of cases are activated mid-outbreak to evaluate whether such a response, without other countermeasures, can help bring an end to the outbreak and provide an estimate to the epidemic duration before the number of cases can be brought to zero.

## 2 Methods

We use a stochastic individual-level branching process model of disease spread in the population. We assume a single index case initiates the outbreak and that transmission in the community is defined by a Negative Binomial offspring distribution with a mean equal to the basic reproduction number ( $R_0$ ), and a dispersion parameter ( $k$ ) to define the individual-level variability in transmission. At present there are no precise estimates for the  $R_0$  for the current Bundibugyo outbreak. We set  $R_0$  to 2.0, 3.0 and 4.0, based on recent growth in suspected and confirmed cases (Institut National de Santé Publique de la République Démocratique du Congo, 2026), to explore the sensitivity of our results to assumed Ebola transmissibility. There is little evidence to inform the individual-level heterogeneity in transmission for Bundibugyo virus. A systematic review of Ebola virus disease found that most estimates show moderate to high heterogeneity ( $k < 1$ ) (Nash et al., 2024). We use the estimate of Polonsky et al. (2021) from the 2018–2020 Ebola epidemic in the DRC ( $k = 0.27$ ).

Ebola cases are isolated after a delay from symptom onset. We model this delay from onset-to-isolation as a Gamma distribution. To evaluate how the speed of isolating cases with a positive test result impacts the ability to bring the outbreak under control, we run scenarios with a mean onset-to-isolation delay of 5 days, 3 days and 1 day (all assuming a standard deviation of 1 day). Cases can be isolated earlier than the delay after symptom onset if they are traced as a contact of a known case (i.e. their infector). When a contact is traced they are isolated as soon as their infector is isolated (or after their delay from symptom onset if earlier). Isolation is assumed to reduce all further transmission ( $R_0^{\text{iso}} = 0$ ). We assess how contact tracing ascertainment influences the probability of bringing the outbreak under control, varying the percentage of contacts traced between 0% and 100% at 20% intervals.

To simulate the speed of transmission the model samples the generation time by first sampling each case’s symptom onset time from the incubation period, then drawing the infection time of each secondary case from a skew-normal distribution centred on the infector’s symptom onset and parameterised by the proportion of transmission that occurs before symptom onset (Hellewell et al., 2020). We use an incubation period estimated from the 2014 Ebola outbreak in West Africa (Gamma distribution, mean = 10.3 days, SD = 8.2 days) (WHO Ebola Response Team et al., 2015). This is similar to the reported incubation period of the Bundibugyo virus (2-21 days, (Coalition for Epidemic Preparedness Innovations, 2026)). We assume that 1% of Bundibugyo virus transmission events occur prior to symptom onset, which is consistent with previous Ebola outbreaks (Centers for Disease Control and Prevention, 2026). Our epidemic model assumes a symptom-based isolation and contact tracing system. Therefore, asymptomatic cases are assumed to be missed by the intervention and continue to transmit in the community with the same transmissibility as symptomatic cases. We assume 1% of Bundibugyo cases are asymptomatic. Again following evidence from past Ebola epidemics which found little evidence of asymptomatic or paucisymptomatic cases (Mbala et al., 2017).

To simulate scenarios consistent with the ongoing Bundibugyo virus outbreak in the DRC, we conditioned each simulation (single index case) on having a cumulative case count in [440, 2230] in at least one of weeks 5, 6 or 7 of the outbreak. Conditioning was implemented by

rejection sampling: branching-process simulations were re-run with independent random seeds until each replicate produced a trajectory meeting the criterion, with up to 1,000 attempts permitted per accepted replicate. We retained 200 accepted replicates per scenario. We assume that no PHSMs (i.e. isolation of cases or contact tracing) are active for the first 42 days of the outbreak (from the spillover infection). After day 42 we assume that individuals that are symptomatic or contact traced from a symptomatic individual are tested and isolated if they receive a positive test result to prevent further infections. We simulate the outbreak for a maximum of 2 years (730 days).

We assume a test sensitivity of 75%. There are no diagnostic assays specifically developed or validated for Bundibugyo virus; rapid antigen tests and the most widely deployed PCR-based assays were developed and validated primarily against Zaire ebolavirus (Broadhurst et al., 2016). We therefore assume an imperfect sensitivity of 75% to reflect the expectation that a proportion of true Bundibugyo cases will return false-negative results when tested with assays optimised for other Ebolavirus species. Only symptomatic cases are tested. Cases returning a false-negative result are not isolated and continue to transmit in the community.

We define outbreak control as no new cases of Ebola virus disease from a certain date to the end of the outbreak. This is equivalent to extinction as we only have a single introduction of the virus into the human population with no subsequent spillover events. In our simulation setup we assume that if the cumulative number of infections exceeds 10,000 then that specific outbreak is considered uncontrollable. We test for outbreak control (extinction) at 3, 6 and 12 months from the activation of the PHSMs.

### 3 Results

The  $R_0$  for transmission in the community impacts the ability to control this Bundibugyo Ebola epidemic (Figure 1). There is a remote chance ( $< 5\%$ ) of ending the outbreak within 3 months from the activation of PHSMs, namely isolation of test-positive cases and contact tracing. Only in the scenario where  $R_0 = 2$  and 100% of contacts for infectious individuals are traced does the outbreak have a highly unlikely ( $> 15\%$ ) chance of ending (extinction) within 3 months with NPIs (Figure 1).

If  $R_0$  is around or below 2 then contact tracing at least 80% of contacts shows a realistic to highly likely possibility of outbreak control within 6–12 months of PHSMs being active (Figure 1). If community  $R_0$  is 3 then 100% of contacts need to be traced to have more than a remote chance of control, and even with 12 months of interventions, there is less than a 50% chance of control. A community  $R_0$  of 4, with no PHSMs for the first 42 days, shows a remote chance of the outbreak ending within a year (Figure 1).

## Proportion of BVD outbreaks controlled

Proportion of outbreaks controlled within 3, 6 or 12 months from the start of PHSM

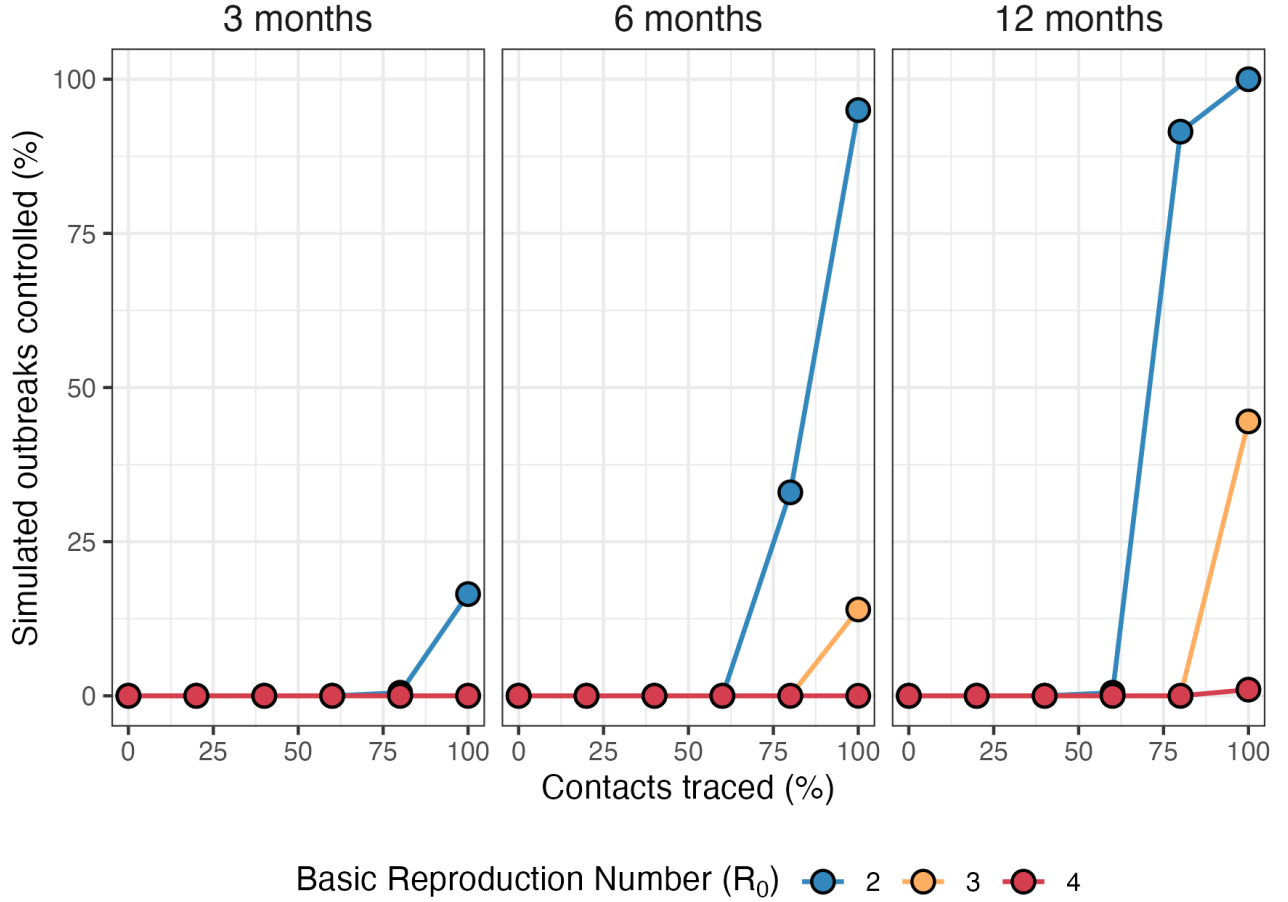


Figure 1: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months of PHSM activation, as a function of the percentage of infectious contacts traced (0%, 20%, 40%, 60%, 80%, 100%). Each point is the proportion of 100 conditioned simulations meeting the control criterion. PHSM activation is modelled as day 42 of the outbreak. The mean onset-to-isolation delay is fixed at 3 days (Gamma, SD = 1 day). Colours distinguish basic reproduction numbers  $R_0 = 2.0$ , 3.0 and 4.0.

All results described assume a mean time delay of 3 days from symptom onset to isolation, which includes taking a test and receiving the positive result. Assuming a longer mean delay to isolating cases of 5 days does not drastically alter the results of the 3-day delay, with probabilities of control marginally worse (for all scenarios not at 0% or 100% of outbreaks controlled) (Figure 2). If cases can be isolated on average 1 day after symptom onset and  $R_0 \sim 2$ , outbreak control is likely within a year if at least 20% of contacts are successfully traced. Outbreak control within 12 months is highly likely if contact tracing can ascertain > 40% of contacts. When  $R_0 > 2$  contact tracing needs to be comprehensive (close to 100%) for control within a year, and even then, the chance of control within 6 months is unlikely to remote (Figure 2).

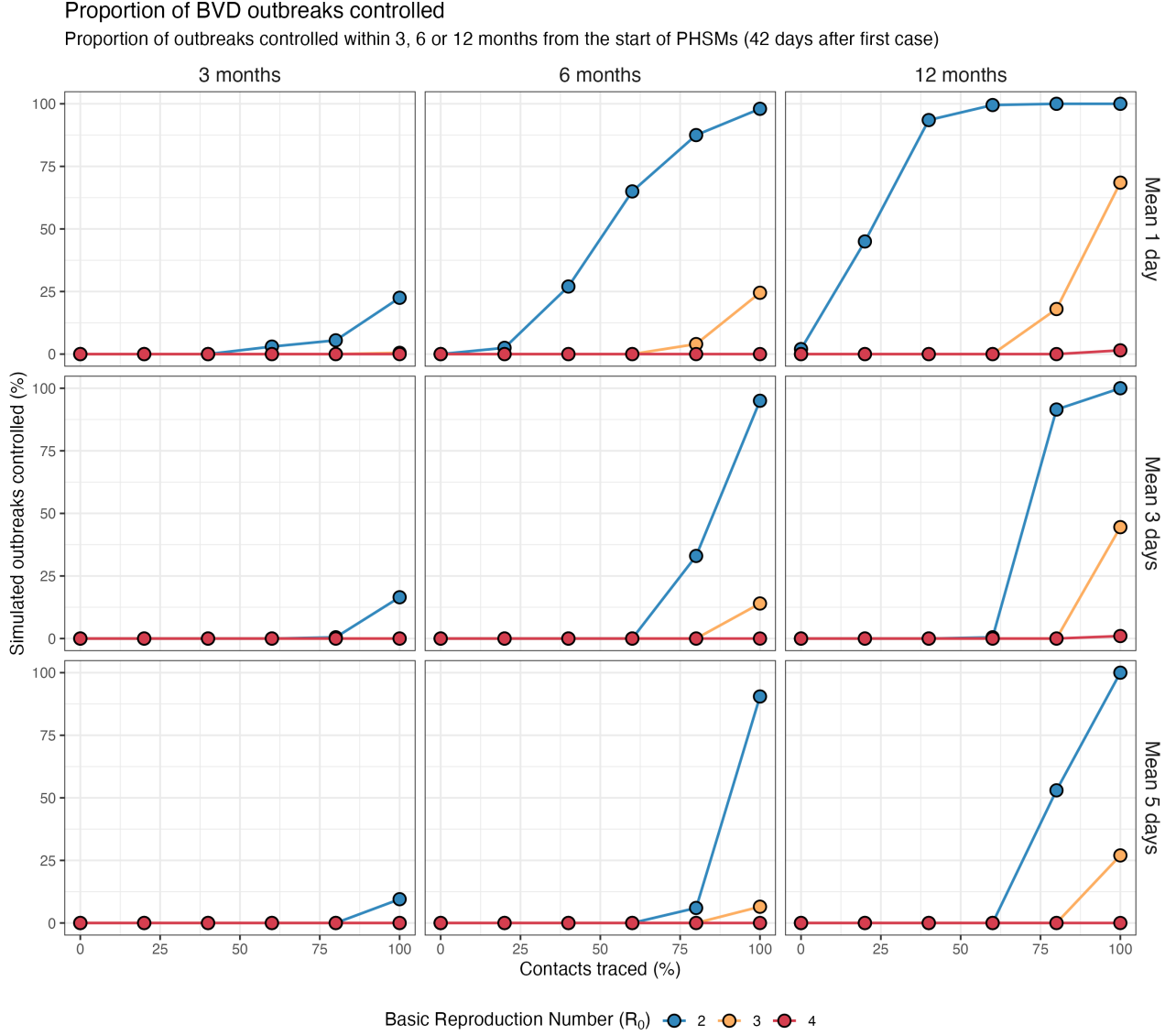


Figure 2: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months (columns) of PHSM activation, as a function of the percentage of infectious contacts traced. Rows show three mean onset-to-isolation delays (top to bottom: 1, 3 and 5 days; Gamma, SD = 1 day). Each point is the proportion of 100 conditioned simulations meeting the control criterion. PHSM activation is modelled as day 42 of the outbreak. Colours distinguish basic reproduction numbers  $R_0 = 2.0, 3.0$  and  $4.0$ .

The descriptions of probability in this report correspond to those in the UK Government Probability Yardstick (Professional Head of Intelligence Analysis, 2025).

## 4 Discussion

- Information on the transmission network would allow a better estimate of the individual-level heterogeneity in transmission (e.g. Althaus (2015)). If transmission is more homogeneous between people in the community then, theoretically, the outbreak would be less controllable (Lloyd-Smith et al., 2005).
- Given the isolation and contact tracing system we model assumes a positive test result to isolate and we assume a test sensitivity of 75% once the intervention is active, test-

ing effectiveness of outbreak control with different sensitivity profile will be important. We could also model the improvement of test sensitivity through the outbreak as tests designed for Bundibugyo become available.

- We do not model capacity constraints on testing, isolating and contact tracing. Each of these are resource-limited aspects and if the number of suspected cases continues to rise these may become overwhelmed, reducing the efficacy of the intervention, or precipitate moving to exposure-based quarantine that does not require a positive test for a contact to isolate once traced.
- Our model does not account for the geographic spread of the Bundibugyo virus, and any geographical variation in response speed or capacity. Our analysis aims at informing what targeted Public Health and Social Measures would be required to bring the outbreak to an end, and what outbreak duration is likely under different epidemiological characteristics until more precise estimates of parameters such as the basic reproduction number or the incubation period for Bundibugyo become available.
- By incorporating the delay to activation of non-pharmaceutical measures we hope to more accurately estimate what control measures are effective and feasible for this outbreak, rather than assuming the intervention is activated by the index case, which would be the case if the outbreak were known and imported into a different region.

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## 5 Code availability

The code to reproduce this analysis can be found at: <https://github.com/joshwlambert/BundibugyoNPI2026>.

The outbreak model code can be found in the R package `ringbp` at <https://github.com/joshwlambert/ringbp@develop>. This analysis used the `develop` branch of the `ringbp` package which can be installed with:

```
remotes::install_github("joshwlambert/ringbp@develop")
```

## 6 AI Disclaimer

This work was assisted by the use of Claude Opus 4.7.